HARSHITA POOJARY

harshita1.poojary@gmail.com | 213-234-8695 | harshitapoojary | HarshitaDPoojary | HarshitaP | Publications

EDUCATION

University of Southern California, Los Angeles, CA.
Master of Science - Computer Science **GPA: 3.81/4.00**
Coursework: Analysis of Algorithms, Deep Learning, Artificial Intelligence, Machine Learning, Natural Language Processing

University of Mumbai, Mumbai, India.
Bachelor of Engineering - Computer Engineering **GPA: 3.83/4.00**
Coursework: Data structures and Algorithms, Soft Computing, Distributed Databases, Web Technologies

TECHNICAL SKILLS

Core Programming Languages: Python
Other Programming Languages: JavaScript, TypeScript, Go, C++, Java, HTML/CSS, Scala
Frameworks & Libraries: Node.js, AngularJS, PySpark, PyTorch, Keras, TensorFlow, OpenCV, Streamlit
Databases: MongoDB, Redis, MySQL, ElasticSearch, Milvus, Hadoop, DynamoDB
Tools & Platforms: Docker, Kubernetes, AWS, Azure, Jenkins, Git, Jira, Postman, Swagger, PowerBI, Tableau, Grafana, Prometheus, Kibana, ELK Stack, OpenTelemetry

WORK EXPERIENCE

AI Intern **Jul 2025 – Present**
Deep Defense Solutions (Remote) *USA*
- Developing automated incident triage by computing severity and **hoax probability** via **Python pipelines** on **AWS**, integrating LLM-derived features with hybrid ML models and embedding-based matching with **ElasticSearch** to historical events at the same geofenced location.

Data Science Intern **Jun 2024 – Aug 2024**
Zero-True (Remote) *New York, USA*
- Delivered **10 interactive demos** showcasing notebook-based ML applications by designing **SQL/Python-integrated dashboards**, enabling stakeholders to dynamically analyze datasets.
- Collaborated with **founders and stakeholders** to evaluate customer and developer feedback, prioritizing high-impact features that shaped the product roadmap and improved developer adoption.

Machine Learning Engineer **Dec 2021 – Jul 2023**
Reliance Jio *Mumbai, India*
- Reduced face registration latency by **40%** by deploying RetinaFace models with **TensorRT FP16** model compression using **Python** and **C++** on **T4/A100 GPUs**, achieving real-time inference across 2 distributed clusters.
- Automated lifecycle deployment workflows across three ML systems using **Python, Bash, Docker, Kubernetes**, and **GitLab CI**, increasing deployment velocity by **50%** while embedding software test engineering practices including unit tests, integration tests, and health checks.
- Troubleshot and resolved on-call incidents by analyzing system metrics, distributed traces (**OpenTelemetry**), logs, and infra bottlenecks, improving **MTTR** by **20%** and ensuring **99%** service uptime.
- Achieved **90% spoof-detection accuracy** and scaled deployments to **20+ sites** by fine-tuning **ResNet-32** on the **H2O.ai** low-code platform, integrating **FastAPI** endpoints and real-time monitoring to track error rates and model drift.
- Led design reviews and code reviews for 4 interns by evaluating system design trade-offs, validating implementation correctness, and aligning on coding standards for production-ready, maintainable software.

Software Engineer **Jun 2019 – Dec 2021**
Reliance Jio *Mumbai, India*
- Engineered distributed microservices & ingestion pipelines (**Node.js, Go, SQL, Kubernetes** on **Linux**) handling **100M+** users, boosting throughput by **35%** across media delivery platforms.
- Built large-scale article/video ingestion in **Node.js** and **SQL** deployed via **Kubernetes** on **Linux systems**, boosting throughput and aggregation efficiency by **35%** across multi-device delivery.
- Improved large-scale system reliability by triaging production issues and performing system diagnosis and resolution using logs, metrics, and distributed traces ***(OpenTelemetry)***, reducing **MTTR** by **20%** across Kubernetes-based services serving 100M+ users.
- Collaborated with PMs and frontend teams to refine content-delivery flows, driving a **12%** increase in engagement.

Software Engineer **Jan 2019 – Jun 2019**
TaksyKraft (Remote) *Hyderabad, India*
- Developed real-time engagement prediction APIs in **Python** and **SQL**, improving targeting accuracy and increasing client ROI by **10%**.
- Shortened reporting cycles by **40%** by building **ETL pipelines** and **Tableau dashboards**, accelerating enterprise decision-making.

PROJECTS

Unified Semantic Space for Multimodal Retrieval (PyTorch, Python, LanceDB)
- Designed a multimodal retrieval framework aligning **BridgeTower** text–image embeddings for video-based Q&A, integrating **LanceDB** for sub-second vector search and deploying an interactive **Streamlit** interface.

Analyzing NLP Techniques on Low-Resource Languages (PyTorch)
- Performed **idiom and metaphor classification** for low-resource **Konkani** using **BERT-based transformers** and applied **attention-head pruning**, maintaining model accuracy while reducing computational overhead for efficient deployment.

Comparison of Model Performance with Knowledge Distillation and Quantization (PyTorch, Python)
- Fine-tuned **LLaMA-3-8B** with **4-bit** quantization and **LoRA**-based logit distillation, reaching **76.9% accuracy** on **GSM8K (8-shot)** while reducing inference cost compared to the original 8B model.